

Updated Business Case for Sub-Project

Sub-Project Name: Member Registration and Authentication System

1.0 Introduction/ Background

The Member Registration and Authentication System is a key component of the Digital Library Management System for the Springfield Public Library , which enables residents to apply for library membership online and securely log in to access digital services. The system was implemented using Firebase Firestore for secure data storage and authentication, and now includes integration with the library dashboard.

2.0 Business Objective

The main objective of this sub-project is to design and implement a secure, efficient, and user-friendly online registration and authentication platform for library members.

3.0 Current Situation and Problem/Opportunity Statement

The existing member registration process is paper-based and time-consuming, requiring residents to visit the library in person. The issues with this are centered around the limitations in accessibility and the additional workload on library staff having to perform manual data entry and verification for every registration.

Opportunities for improvement include:

- Transitioning to an online system to reduce administrative overhead. - Enhance member experience with 24/7 access to registration and account management.
- Strengthens data integrity and improves security through authentication protocols.
- The new system also supports navigation to a separate dashboard system developed by another team

4.0 Critical Assumption and Constraints

Assumptions:

- Stable internet access is available to all residents.
- Admins/Librarians have access to the staff dashboard for approvals.
- Security verification and privacy standards will be met according to institutional IT policy.
- We assume Firebase services remain stable and accessible during operation.

Constraints:

- The project must be completed within **2 months** and under a **\$160,000 budget**.
- Integration must align with the architecture of other sub-projects.
- The project required alignment with another team's dashboard timeline, which affected integration timing.

5.0 Analysis of Options and Recommendation

Option 1: Semi-Automated system

Online form with manual processing - Pros: Moderate cost and easy transition. Option 2: Fully Online system

Automated submissions, librarian approval, and PIN-based login. High efficiency.

6.0 Preliminary Project Requirements

- Online member application form with validation.
- Secure authentication system (library card number + PIN).
- Database integration with central member accounts.
- API integration for Staff Dashboard (Sub-project 7).
- Integration with Firebase Firestore for real-time data storage and authentication.

7.0 Budget Estimate and Financial Analysis

DESCRIPTION COST

Software Development and Testing \$90,000

Security and Authentication Modules \$25,000

Server Hosting and Database Setup \$15,000

Contingency \$30,000

TOTAL \$160,000

Financial Analysis: This investment is expected to reduce registration processing time by over 70% and lower administrative costs by at least 25% annually. Improved user experience is projected to increase active membership registrations by 40% within the first year of Integration.

8.0 Schedule Estimate

PHASE ACTIVITIES DURATION Planning Research, requirements gathering and designing 1 week.

Development Build the system. 4 weeks Integration Connect with database and Staff 1 week

Testing Functional and Security testing 1 week

Deployment Final deployment 1 week

Actual development included small delays due to Firebase setup and dashboard integration.

9.0 Potential Risks

Risk 1: Data Breaches (High Risk)

Risk 2: Integration delays due to sub-projects (Low Risk)

Risk 3: Inadequate Budget (Low Risk)

Risk 4: Firebase configuration or authentication errors (Medium Risk).

10.0 Exhibits

Exhibit A: Financial Analysis

-See Section 7 for detailed breakdown on project costs.